Data Science and Analytics, B.S.

Location(s): On Campus

PROGRAM DESCRIPTION | PROGRAM STUDENT LEARNING OUTCOMES | ADMISSIONS, ENROLLMENT, & GRADUATION CRITERIA | PROGRAM COURSE REQUIREMENTS | HAVE QUESTION CONTACT US!

Program Description

The Bachelor of Science with a major in Data Science and Analytics will provide a student with foundational mathematical, statistical, and computational knowledge, skills, and methodologies within the context of the ethical and professional standards of Data Science. A st also complete at least 16 hours of courses in either a domain of expertise in data science and analytics or a minor to provide them a context in which to apply their data science abilities. Thus, the degree will enable the student to either begin a career in industry, government community and non-profit organizations in a range of domains, or pursue graduate study.

Students will begin the program by building a foundation in mathematics, statistics, computer programming, and algorithmic techniques. They will then take 38 credit hours of data science core courses covering the fundamentals of data science, programming, machine lea mining, data science ethics, and communication. After completing the core, students will complete 6 credit hours of elective courses in data science and statistical learning. Students will also be required to take at least 16 hours in a suitable domain knowledge concentration exploring an expert area of application. The program will conclude with a required data science capstone course, in which the student will demonstrate overall knowledge of the discipline by completing a data science project, incorporating all the knowledge learned in the

Program Student Learning Outcomes

Students who successfully complete this program will be able to:

1. Analyze a complex computing problem and apply principles of computing and other relevant disciplines to identify solutions.
2. Design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of the program's discipline.
3. Communicate effectively in a variety of professional contexts.
4. Recognize professional responsibilities and make informed judgments in computing practice based on legal and ethical principles.
5. Function effectively as a member or leader of a team engaged in activities appropriate to the program's discipline.
6. Apply theory, techniques, and tools throughout the data science lifecycle and employ the resulting knowledge to satisfy stakeholders' needs.


This program is a part of the College of Computing and Software Engineering.

Admissions, Enrollment, and Graduation Criteria

Admissions Criteria

Admission to this program is open to all students who meet Kennesaw State University's general admission standards. Visit the Admissions section of the Catalog for more details.

Enrollment Criteria

This program does not have specific enrollment criteria; however, students are expected to meet the requirements of Academic Policy 4.0 ACADEMIC STANDING, DISMISSAL, & REINSTATEMENT.

Graduation Criteria

Each student is expected to meet the requirements outlined in Academic Policy 5.0 PROGRAM REQUIREMENTS & GRADUATION.

Program Course Requirements

Core IMPACTS Curriculum (42 Credit Hours)

KSU's General Education Core IMPACTS Curriculum

Core IMPACTS Curriculum Specific to this Major

Students should take MATH 1113 or higher in Mathematics & Quantitative Skills and MATH 1179 or higher in Applied Math. Other options are available, please see an advisor for details; however, selection of other options in IMPACTS may lengthen time to gradua

Core Field of Study (18 Credit Hours)

Students must earn a grade of "C" or better in these courses.

MATH 2202: Calculus II

MATH 2202: Calculus II

4 Credit Hours

Prerequisite: MATH 1190 or (MATH 1179 and MATH 1189)

Description

This course is the second in the calculus curriculum and consists of two parts. The first part is concerned with the techniques of integration and applications of the integral. The second part is concerned with infinite sequences and series.

Course Learning Outcomes

Students who successfully complete this course will be able to:

1. Evaluate a wide variety of standard integrals.
2. Apply integrals to a variety of problems involving area, volume, curve length, work done by a force and other similar quantities.
3. Understand the concept of convergence of improper integrals and be able to evaluate some basic improper integrals;
4. Understand the concept of convergence of infinite series and be able to derive and use a variety of tests for convergence of a series;
5. Understand the concept of a power series, be familiar with the important properties of power series, and understand the concept of the Taylor series of common functions.

CSE 2300: Discrete Structures for Computing

CSE 2300: Discrete Structures for Computing

3 Credit Hours

Prerequisite: ((CSE 1321 and CSE 1321L with a grade of "B" or higher) and (MATH 1113 or MATH 1190 or MATH 1179)

Description

Coverage of discrete structures is crucial to any program in computing. This course covers propositional and predicate logic, proofs, set theory, relations and functions, algorithms and complexity theory, matrices, graphs and trees, and combinatorics. Throughou emphasis will be on applications of these concepts in computing.

or

MATH 2345: Discrete Mathematics

MATH 2345: Discrete Mathematics

3 Credit Hours

Prerequisite: MATH 1113 or MATH 1190 or MATH 1179

Description

An introduction to the fundamentals of discrete mathematics. Topics include sets, formal logic, methods of proof, counting relations, functions, graphs and trees, and finite state automata.

Notes: Not intended for mathematics or mathematics education majors.

STAT 1401: Elementary Statistics

STAT 1401: Elementary Statistics

3 Credit Hours

Description
This is a non-calculus based introduction to statistics. Course content includes descriptive statistics, probability theory, confidence intervals, hypothesis testing, and other selected statistical topics.

Course Learning Outcomes
Students who successfully complete this course will be able to:

1. Use statistical vocabulary and notation appropriately.
2. Identify appropriate methods for collecting data.
3. Distinguish the difference between qualitative and quantitative data and recognize when each is appropriate.
4. Describe and graphically represent statistical data.
5. Correctly interpret statistical graphical displays.
6. Identify measures of center and variation and use them appropriately to describe distributions.
7. Compute basic probabilities and correctly use computations for application.
8. Build confidence intervals to estimate population parameters such as means and proportions from statistical data.
9. Perform hypothesis tests for population parameters and appropriately interpret the results.
10. Successfully use technology to describe, analyze, and perform inferential statistics.
11. Apply statistical concepts to real-life scenarios.

or

STAT 2332: Probability and Data Analysis

STAT 2332: Probability and Data Analysis

3 Credit Hours

Prerequisite: MATH 1190 or MATH 1179
Description
This course is a foundational, calculus-based introduction to statistics and probability. The following conceptual themes will be developed through the process of statistical investigation: exploratory data analysis (univariate and bivariate), fundamentals of exper design and sampling, planning and conducting a study, exploring random phenomenon using probability and simulation, and the fundamentals of statistical inference. Technology is integrated into each theme, and the statistical software package used will be cl the instructor.

CSE 1321: Programming and Problem Solving I

CSE 1321: Programming and Problem Solving I

3 Credit Hours

Concurrent: CSE 1321L
Description
This course provides an introduction to computing with a focus on programming. Instruction centers on an overview of programming, problem-solving, and algorithm development. Particular topics include object-oriented design/programming, primitive data arithmetic and logical operators, selection and repetition structures, interactive user input, exception handling, using and designing basic classes, single-dimensional data structures with searching and sorting, and arrays. Programming assignments focus on tech good programming style including proper documentation. The student is taught to efficiently design, code, and debug problem solutions and the relationship between correct code and security.

Course Learning Outcomes
Students who successfully complete this course will be able to:

1. Apply computational methods to analyze data and solve real-world problems.
2. Demonstrate the appropriate use of data types and logical, relational, and arithmetic expressions in programs.
3. Construct algorithmic solutions that incorporate selection and repetition.
4. Implement collections appropriately to store and analyze data.
5. Define functions and classes in program solutions.

CSE 1321L: Programming and Problem Solving I Laboratory

CSE 1321L: Programming and Problem Solving I Laboratory

1 Credit Hours

Concurrent: CSE 1321
Description
Lab activities with programming to accompany CSE 1321.

Course Learning Outcomes
Students who successfully complete this course will be able to:

1. Apply computational methods to analyze data and solve real-world problems.
2. Demonstrate knowledge of the syntax of a specific programming language.
3. Employ IDEs and compilers to learn how to compile, debug, run and test programs.
4. Demonstrate the appropriate use of data types and logical, relational, and arithmetic expressions in programs.
5. Construct algorithmic solutions that incorporate selection and repetition.
6. Implement collections appropriately to store and analyze data.
7. Define functions and classes in program solutions.

CSE 1322: Programming and Problem Solving II

CSE 1322: Programming and Problem Solving II

3 Credit Hours

Prerequisite: CSE 1321 and CSE 1321L
Concurrent: CSE 1322L and (MATH 1113 or MATH 1190 or MATH 1179 or MATH 2202)
Description
The second course in computing provides coverage of more advanced topics of object-oriented programming. This includes the use of static variables and classes, non-linear data structures, inheritance and polymorphism, file input/output, exception handling, and parameterized types. Elementary data structures (linked lists, stacks, and queues) are introduced to solve application problems. Graphical user interfaces, parallel programming, database programming, and event-driven programming are also introduced. S will use good programming style including proper documentation.

CSE 1322L: Programming and Problem Solving II Laboratory

CSE 1322L: Programming and Problem Solving II Laboratory

1 Credit Hours

Prerequisite: CSE 1321 and CSE 1321L
Concurrent: CSE 1322
Description
This course is the required and supervised lab course to accompany CSE 1322.

Major Requirements (37 Credit Hours)

Students must earn a grade of "C" or better in these courses.

DATA 3010: Computer Applications of Statistics

DATA 3010: Computer Applications of Statistics

3 Credit Hours

Prerequisite: STAT 1401 or ECON 2300 or STAT 3125 or STAT 2332 or MATH 3332

Description

This is an intermediate survey course of computer-based statistical software applications in the analysis and interpretation of data. Topics include developing a proficiency in coding in multiple languages through quantitative applications. Software packages inc most in-demand statistical languages and packages in the marketplace. (e.g. Python, SAS, R)

Course Learning Outcomes

Students who successfully complete this course will be able to:

1. Function as an informed user and critical consumer of statistics able to answer questions in an intelligent, informed, and insightful manner and pose intelligent questions concerning data, data analysis, and software use.
2. Obtain and correctly interpret the measures of central tendency and dispersion of a data set.
3. Generate and correctly interpret a variety of graphics for both univariate and multivariate analyses.
4. Generate random numbers and use them to perform Simple Random Sampling.
5. Generate and correctly interpret confidence intervals.
6. Represent themselves as proficient in each of these packages in a job interview or professional environment.
7. Identify types of statistical variables.
8. Differentiate the roles of explanatory and response variables.
9. State a proper research hypothesis.
10. Choose an appropriate statistical analysis for a given situation.
11. Summarize and communicate analysis results through a professional document.

STAT 3120: Statistical Methods I

STAT 3120: Statistical Methods I

3 Credit Hours

Prerequisite: STAT 1401 or DATA 1501 or STAT 2332 or MATH 3332

Description

This course is designed to provide students with a foundation in statistical methods, including confidence intervals for population parameters, correlation, simple linear regression and hypothesis testing (F and T-tests for regression, chi-square for independence and paired sample T-tests). These concepts are taught with heavy emphasis on statistical coding software and real-world datasets from a variety of disciplines.

or

STAT 3125: Biostatistics

STAT 3125: Biostatistics

3 Credit Hours

Prerequisite: BIOL 1107 or BIOL 1108 or CHEM 1212 or permission of the instructor

Description

In this course students use descriptive statistics and visual displays to describe biological and medical data. They perform and analyze results of statistical analyses which may include confidence intervals, correlation, linear regression, odds/risk ratios, and hypo testing (Chi-square for independence, 2 group and paired sample t-tests). Analyses are performed using the statistical software R.

STAT 3130: Statistical Methods II

STAT 3130: Statistical Methods II

3 Credit Hours

Prerequisite: DATA 3010 and (STAT 3120 or STAT 3125 or STAT 2332 or PSYC 3000)
Description
Students continue to build their foundation in statistical methods in this course. They will conduct non-parametric methods (Wilcoxon Signed Rank, Rank Sum, and Kruskal Wallis tests), ANOVA and multiple regression. These concepts are taught with heavy er
statistical coding software and real-world datasets.

DATA 3230: Data Visualization

DATA 3230: Data Visualization

3 Credit Hours

Prerequisite: STAT 1401 or DATA 1501 or STAT 2332 or STAT 3125
Description
This course introduces students to the field of data visualization. The course covers basic design and evaluation principles to prepare and analyze large datasets, and standard visualization techniques for different types of data using modern data visualization so
The course prepares students to communicate clearly, efficiently, and in a visually compelling manner to a variety of audiences.

MATH 3260: Linear Algebra I

MATH 3260: Linear Algebra I

3 Credit Hours

Prerequisite: MATH 1190 or (MATH 1179 and MATH 1189)
Description
An introduction to linear algebra and some of its classical and modern applications. Among topics to be included will be systems of linear equations, matrices, determinants of matrices and applications, vector spaces, and inner product spaces. Significant use of
technology will be employed in performing matrix computations.

DATA 3300: Data Science Ethics

DATA 3300: Data Science Ethics

3 Credit Hours

Prerequisite: STAT 3130
Description
As the field of data science and artificial intelligence continues to rapidly grow, so does the need for strong ethical guidelines. Throughout this course, students will learn the foundational ethical theories and frameworks, and the origins of ethics within data scie
Students will use case studies to learn about the ethical dilemmas around the collection, management, and use of data, the use of models and algorithms, and the future of artificial intelligence and machine learning. Topics include Privacy, Informed Consent, Ov
Security, Bias, Misinformation, Data Governance and Codes of Ethics.

CSE 3153: Database Systems

CSE 3153: Database Systems

3 Credit Hours

Prerequisite: (CSE 1322 and CSE 1322L) or IT 1113 or (IT 1114 and IT 1114L)
Description
The topics in this course span from a review of the traditional file processing systems to database management systems. Topics include files systems and file processing logic, planning, and major phases of database development: analysis, design and implementa use an SQL based database product such as Oracle.

or

CS 3410: Introduction to Database Systems

CS 3410: Introduction to Database Systems

3 Credit Hours

Prerequisite: CSE 1322 and CSE 1322L
Description
Introduction to database management systems, database processing, data modeling, database design, development, and implementation. Particular emphasis is placed on the relational approach to database management and processing, which focuses more on t nature of a database than its physical characteristics. Relational database programming assignments are drawn from the fields of business. Includes implementation of current DBMS tools and SQL. Ethical and security topics related to databases will be introduc

DATA 4000: Data Science Communication

DATA 4000: Data Science Communication

3 Credit Hours

Prerequisite: STAT 2332 or (DATA 3010 and (STAT 3120 or STAT 3125))
Description
This course equips students to orally communicate data analysis results adapted to both technical and non-technical audiences. Students learn and practice essential data presentation skills, such as using narratives and visuals to communicate data analysis insig solving business problems.

DATA 4030: Programming in R

DATA 4030: Programming in R

3 Credit Hours

Prerequisite: DATA 3010 or STAT 3125
Description
In this course, students will learn R programming to effectively manage, explore, visualize, and present data. The course covers practical statistical-computing issues, including reading, manipulating, and analyzing data, using control statements, existing functi user-created functions. Reporting results using R packages, such as R Markdown, may also be covered.

or

DATA 4140: Python for Data Science

DATA 4140: Python for Data Science

3 Credit Hours

Prerequisite: DATA 3010 and (STAT 3130 or IET 3403 or ISYE 3600)
Description
This course introduces students to analytic methods using Python. The core focus is the development of Python knowledge within an analytic model development focus. Students will learn fundamental data structures, key algorithms and their application in ap

analytic/machine learning methodologies.

STAT 4210: Applied Regression Analysis

STAT 4210: Applied Regression Analysis

3 Credit Hours

Prerequisite: STAT 3130
Description
This course is designed to provide students with various regression procedural methods, including simple linear and multiple regression models. Students will diagnose multicollinearity, identify outliers and influential observations, and assess assumptions to c
validate models. Other topics will include parameters inferences, variable transformations, and qualitative predictors.

DATA 4310: Statistical Data Mining

DATA 4310: Statistical Data Mining

3 Credit Hours

Prerequisite: STAT 3130 or permission of the instructor.
Description
Data Mining is an information extraction activity whose goal is to discover hidden facts contained in databases, perform prediction and forecasting, and generally improve their performance through interaction with data. The process includes data selection, clea
coding, using different statistical, pattern recognition and machine learning techniques, and reporting and visualization of the generated structures. The course will cover all these issues and will illustrate the whole process by examples of practical applications.
students will use recent SAS Enterprise Miner software.

DATA 4990: Data Science Capstone

DATA 4990: Data Science Capstone

3 Credit Hours

Prerequisite: STAT 4210
Description
Capstone projects challenge students to acquire and analyze data to solve real-world problems. Students will have to synthesize and strengthen the knowledge and skills learned through the program such as data visualization, inference and modeling, data wra
data organization, data mining and machine learning, as well as storytelling with the data.

One (1) credit hour carried over from Applied Math or one (1) credit hour of university electives (1000-4000 level coursework from the University Catalog).

Major Electives (6 Credit Hours)

Students must earn a grade of "C" or better in these courses. Select six credit hours from the following list of courses:

DATA 3396: Cooperative Study

DATA 3396: Cooperative Study

1-3 Credit Hours

Prerequisite: Approval of the coordinator of cooperative education/internship.
Description
This course is a supervised work experience program for a minimum of two academic semesters at a site in business, industry, or government. It is for sophomore, junior, or senior-level students who wish to obtain successive on-the-job experience in conjunctior

their academic training.

DATA 3398: Internship

DATA 3398: Internship

1-9 Credit Hours

Prerequisite: Approval of the program coordinator and department chair.
Description
This course is a supervised, credit-earning work experience of one academic semester with a previously approved business firm, private agency, or government agency.

CSE 4983: CSE Computing Internship

CSE 4983: CSE Computing Internship

3 Credit Hours

Prerequisite: Senior standing
Description
This course helps students gain practical experience through real-world projects and professional work. Students demonstrate an ability to apply computing principles and technologies relevant to their major in a specific real-world project jointly supervised by industry mentor and a faculty advisor. Students work in a project team in an enterprise environment demonstrating ethical behavior as a computing professional, an understanding of social, professional and ethical issues related to computing, and an ability to i the knowledge acquired in preceding courses. Communication skills and leadership are also evaluated as well as professional computing skills and knowledge. 150+ hours per semester required at an internship site. The course can not be repeated for credit.

or Any DATA or STAT 4000-level course

University Electives (17 Credit Hours)

In accordance with KSU Graduation Policy, students must earn a grade of "D" or better in these courses while maintaining a minimum 2.00 cumulative GPA.

Free Electives (17 Credit Hours)

Select 17 credit hours of 1000-4000 level coursework from the University Catalog. Students are encouraged to take courses that focus on a particular domain with data science applications. These hours can also be used to earn a minor in another discipline.

Program Total (120 Credit Hours)

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